Bridging Universities, Capital, and Real-World Projects

A Translational Framework for Impact Measurement

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Abstract

Universities produce vast knowledge and innovation, yet translating academic outputs into real-world projects and measurable societal impact remains challenging. Existing impact measurement frameworks often overlook early-stage university research, limiting capital engagement.

Here we propose a translation-aware, staged impact measurement framework connecting academic, translational, and societal/capital dimensions. This framework tracks credible signals across the innovation pathway, enabling earlier and more effective collaboration among universities, investors, and project partners.

Case studies from Cambridge and US entrepreneurship accelerators illustrate how applied research aligns with sustainable development goals through spinouts. By providing structured visibility into translational progress, the framework supports strategic investment and policy decisions without compromising academic independence.

This approach reframes impact measurement as a forward-looking coordination mechanism, fostering interdisciplinary collaboration and enhancing the societal relevance of university research.

1. Introduction: The Translation Gap Between Universities and the Real World

Universities play a central role in producing new knowledge, training human capital, and advancing scientific and social understanding. Despite this, a substantial proportion of academic outputs remain confined to academic publications, laboratories, or classrooms, with limited pathways into real-world deployment. At the same time, investors, foundations, and public institutions increasingly seek projects that generate measurable social and environmental impact, yet often struggle to identify investable opportunities emerging from academic environments.

This disconnect is not primarily a failure of innovation or intent, but rather a failure of translation. Academic knowledge is rarely designed to speak the language of capital, implementation, or impact measurement. Conversely, investment and policy systems frequently lack the conceptual tools to engage meaningfully with early-stage academic outputs. As a result, potentially transformative ideas remain underutilized, while capital flows toward projects that are easier to measure rather than those with the greatest long-term potential.

2. The Limits of Existing Impact Measurement Approaches

Over the past decade, impact measurement has become a central concern in policy, philanthropy, and investment. Numerous frameworks have been developed to assess social and environmental outcomes, ranging from high-level goal alignment to granular performance metrics. While these approaches have advanced accountability in many sectors, they exhibit structural limitations when applied to university-originated research and teaching outcomes.

- Many widely adopted frameworks are outcome-oriented but not process-sensitive. They tend to evaluate impact only after a project has reached a mature or commercialized stage, leaving early-stage academic work effectively invisible to capital providers.
- Existing models often assume organizational ownership of projects, whereas universities typically produce dispersed outputs—ideas, methods, talent, and prototypes—rather than fully formed ventures.
- Most systems prioritize short-term measurability over long-term transformation, which disadvantages research-driven initiatives whose value emerges over extended time horizons.

3. Universities as Translational Engines, Not Just Knowledge Producers

To address this gap, it is necessary to reconceptualize the role of universities in the impact ecosystem. Rather than viewing universities solely as upstream knowledge producers, this article proposes understanding them as translational engines—institutions capable of moving ideas across domains, provided that appropriate interfaces exist.

University teaching and research generate at least three categories of value:

- **Epistemic value:** new theories, data, and methods
- **Human capital value:** trained individuals capable of applying knowledge in complex contexts
- **Institutional credibility:** legitimacy, trust, and long-term orientation.

3.1 Structural Constraints Within Academia

While universities are increasingly encouraged to demonstrate real-world relevance and societal impact, academic systems remain shaped by structural constraints that limit translational engagement.

First, capacity constraints play a significant role. Reductions in administrative and support staff, combined with increasing teaching, compliance, and funding-related workloads, leave many academics with limited time and institutional support to engage meaningfully in translational initiatives.

Second, incentive misalignment persists within academic career structures. Progression and recognition continue to be heavily tied to publication output and citation metrics, reinforcing a "publish or perish" culture in which translational activities—such as industry collaboration, pilot implementation, or applied experimentation—are often undervalued or rendered invisible within formal evaluation systems.

Third, governance and boundary-setting challenges shape academic attitudes toward external engagement. Commercial agendas may be perceived as compromising research independence, while unclear rules around IP, conflicts of interest, data access, ethics, and timelines can make collaboration feel risky or extractive. Requests for unpaid consultancy or externally defined research agendas can further reduce willingness to participate in industry-led or policy initiatives.

These constraints do not reflect resistance to impact itself. Rather, together they indicate a systemic failure of translation.

4. A Translation-Aware Staged Framework for Impact Measurement

This article proposes a three-layer, translation-aware impact measurement framework designed for university-originated outputs. "Translation-aware" means the framework tracks credible signals across the full pathway—from academic production, through translational initiatives, to downstream societal and capital outcomes—so that different stakeholders can engage earlier with appropriate expectations and evidence.

4.1 Academic Impact

This level captures the intrinsic contributions of universities, including:

- Knowledge creation (publications, methods, datasets)
- Educational outcomes (skills development, interdisciplinary training)
- Research capacity building.

Metrics combine qualitative peer validation with well-established quantitative indicators. In addition to expert review, academic impact can be measured using bibliometrics such as citations, citation-network and co-authorship-network measures, venue quality signals, and open-science reuse. These measures establish the credibility and integrity of the underlying work while remaining explicit about their limits and context.

4.2 Translational Impact

The translational layer represents the critical missing interface. It focuses on:

- Pilot projects and field experimentation
- Cross-sector collaboration (universities with NGOs, hospitals, municipalities, or firms)
- Proof-of-concept validation in real-world contexts.

Measurement at this stage emphasizes learning velocity, adaptability, and contextual relevance, rather than financial returns or large-scale outcomes.

4.3 Societal and Capital Impact

At the final layer, impact becomes legible to broader systems:

- Capital mobilization and leverage
- Cost efficiency or system-level improvements
- Measurable social, health, or environmental outcomes

Importantly, this level builds upon the credibility and evidence accumulated in the earlier stages and interprets outcomes as trajectories over time.

5. Why Translational Measurement Enables Capital Engagement

Without translational impact metrics, capital faces a binary choice: either invest too late, after innovation potential has already narrowed, or avoid academic engagement altogether. By contrast, a staged measurement framework allows different forms of capital—philanthropic, public, and impact-oriented—to participate at appropriate points along the translation pathway.

This approach also reduces risk asymmetry. Universities are no longer required to overpromise outcomes, while investors gain structured visibility into progress without imposing premature performance expectations.

5.1 Methodological Extensions and Implementation Pathways

To make this framework practically useful, future research could explore methodological tools that convert this conceptual model into operational strategies. Several directions are promising:

- **Network and graph-based methods**
- **Algorithmic mapping of institutional collaboration**
- **Impact signal extraction**.

Rather than offering a single solution, this proposal invites multidisciplinary and cross-sector collaboration to iterate, test, and refine the framework. It provides a conceptual backbone that could be adapted into metrics, visualizations, or bespoke decision tools.

5.2 Context-Embedded Impact Measurement and Scorecard Generation

The methodological directions outlined above can be composed into an integrated measurement architecture that moves from bespoke, data-driven impact assessment toward scalable decision tooling. This architecture operates across three layers and, over time, through three stages of maturity.

5.2.1 Three Layers of Context-Embedded Measurement

For each investment or translational project, the measurement system constructs a context-specific impact assessment by integrating three layers:

The definitional layer. Each investment defines what impact means for it specifically—a structured set of impact dimensions, indicators, and thresholds reflecting the particular theory of change for that case.

The data layer. Ecosystem partners—universities, TTOs, accelerators, funds, NGOs, public bodies—continuously produce signals: pilot milestones, partnership announcements, hiring patterns, policy citations, media mentions, clinical trial registrations, patent filings, and community forum discussions.

The network and relational layer. For each investment, a local subgraph is constructed from the broader ecosystem network. Nodes represent the actors and outputs relevant to that investment—the research group, the spinout, co-investors, pilot sites, policy bodies that cited the work—and edges represent the relationships and information flows between them. Impact signals are then derived from the structural and temporal properties of that subgraph, including:

- **Connectivity and reach**—how many distinct sectors or actor types does the investment's translational activity touch.
- **Information flow velocity**—how quickly do signals propagate through the local network, from pilot result to investor awareness to policy uptake.
- **Structural position**—does the investment occupy a bridging position between otherwise disconnected communities (high betweenness centrality), or is it embedded within a dense cluster. Bridging positions often indicate higher translational potential.
- **Temporal evolution**—how is the subgraph changing over time. Growing connectivity, new actor types entering, increasing density—these are leading indicators of translational momentum.

The result is a context-embedded measure: it cannot be divorced from the particular network of relationships and information flows that produced it, and it updates as the ecosystem evolves rather than sitting static in an annual report.

5.2.2 From Accumulation to Typified Scorecards

Over a sufficient number of investments assessed through this system, the accumulated case records—each linking starting conditions, early network signatures, and translational outcomes—constitute a training dataset. This enables a three-stage maturity pathway:

Stage 1: Accumulation and learning. Every investment that passes through the network-embedded measurement system produces a rich case record: the initial impact definition, the subgraph structure and its evolution over time, the data feeds that contributed signals, and the eventual translational outcomes (successful deployment, policy adoption, capital leverage, failure, pivot).

Stage 2: Pattern extraction and typification. From the accumulated dataset, recurring structural profiles can be identified, types of investments that share similar network signatures, impact trajectories, and outcome distributions. These are empirically derived, not hypothetical rules. Crucially, the resulting typification is not generic but archetype-specific: a family of profiles, each characterising a distinct class of translational investment.

Stage 3: Scorecard generation. The typified patterns become the basis for structured scorecards—decision tools that translate the learned network signatures into a set of observable, assessable criteria that a human investor can evaluate without requiring real-time data integration. Each scorecard item is grounded in an empirically observed correlation between that feature and translational outcomes in the accumulated dataset. The scorecard is, in effect, a lossy compression of the network-embedded measurement system into a portable format. The system is designed to be recursive. Investments initially evaluated via scorecard can subsequently be brought into the full data-integrated system, their outcomes tracked, and the scorecards iteratively refined. The static tool is always a derivative of the live system, and it improves as the live system accumulates more cases. The characterised case studies presented in Section 9 illustrate the seed structure for this process: each case, once mapped against structured dimensions, constitutes a labelled example from which typification can begin.

6. Measuring Real-Life Impact in Translational Contexts

While the primary function of translational impact measurement is to enable earlier and more appropriate capital engagement, it is equally important to clarify how real-life impact becomes visible within this framework.

In translational contexts, real-life impact should not be understood as a single measurable outcome, but as a set of observable signals that emerge as academic work interacts with real-world environments. These signals allow capital providers to assess relevance and progress without relying on premature performance metrics.

Three forms of real-life impact observation are particularly salient:

- **Implementation signals:** initiation and continuation of pilot projects, field trials, or institutional collaborations in which academic insights are actively applied.
- **Behavioural and institutional shifts:** changes in organizational behaviour, decision-making processes, or professional practices.
- **Outcome trajectories:** improvements in service delivery, cost efficiency, access, or wellbeing.

Taken together, these observational dimensions provide capital with structured visibility into translational progress, while preserving the exploratory nature of early-stage academic engagement.

7. Data Sources and Collaboration Pathways for Translational Impact

To operationalize the proposed measurement framework, clear sources of data and sustainable collaboration models are essential. However, a critical challenge must be acknowledged at the outset: the vast majority of capital providers—including venture capital, private equity, and institutional investors—currently have little or no awareness of translational impact metrics as described in this framework. Investor decision-making remains overwhelmingly oriented toward commercial traction, revenue growth, and market positioning, with limited visibility into the upstream signals that indicate translational progress. This information asymmetry means that even where high-quality translational data exists, it rarely reaches the actors who could act on it. Bridging this gap requires not only better data, but active mechanisms for surfacing, curating, and circulating that data across stakeholder communities in near real time.

7.1 Translational Impact: Types of Observable Data

Three primary categories of data are envisioned:

- **Translational Activities:** e.g. pilot programs, cross-sector collaborations, early-stage interventions.
- **Psychometric Profiles:** motivations, values, and personality traits associated with impact inclination—particularly useful in understanding founder-investor-academic alignment.
- **Real-World Signals:** such as public launch of spinouts, media mentions, policy references, or stakeholder uptake.

7.2 Sources and Partners

Key data sources and collaboration partners include:

- **VC / PE Funds:** Real deal data across early-stage and impact funds can serve as a proxy for translation maturity, project staging, and capital alignment.
- **Institutions and networks:** University Tech Transfer Offices; Public-sector partners (e.g. NHS, local governments); NGOs involved in field application; Academic networks (e.g., ORCID, LinkedIn, research datasets); Textual analytics from policy documents, media reports, or even AI-enhanced semantic tracking of citation impact.

7.3 Ecosystem and Community as Infrastructure for Data and Coordination

The data categories and partner types outlined above do not, on their own, constitute a functioning measurement system. Their value depends on an active ecosystem—the structured interdependence between universities, investors, government bodies, accelerators, NGOs, and industry partners—functioning not merely as a metaphor but as a system in which each actor's outputs become inputs for others. Without ecosystem-level infrastructure, data remains fragmented across siloed institutions.

Investors' lack of awareness of translational impact metrics is not primarily a knowledge deficit resolvable through education alone. It reflects the absence of community-level mechanisms for surfacing and circulating information that would make translational signals legible and actionable.

Traditional data sources—annual reports, published research, fund performance data—are too slow, too aggregated, and too detached from the translational process to serve this function. What is needed instead is community-sourced information: the practitioner networks, working groups, and cross-sector forums through which translational actors—academics, investors, entrepreneurs, policymakers—share real-time observations, surface emerging opportunities, and collectively interpret signals of progress or failure.

Ecosystem and community thus serve as the essential infrastructure through which the data sources and collaboration pathways described in this section become operational. They also provide the continuous data feeds required by the context-embedded measurement architecture proposed in Section 5.2: without an active ecosystem generating real-time signals, the network and relational layer of that system has no inputs, and the accumulation process from which typified scorecards are derived cannot begin.

7.4 Incentives and Justifications for Collaboration

Building a functional measurement system requires shared incentives. Potential benefits for partners include:

- **For Funds:** A forward-looking tool to evaluate early-stage impact potential beyond commercial traction.
- **For Universities:** Structured visibility into real-world application of academic outputs.
- **For Governments and NGOs:** Alignment with policy mandates, transparency in innovation deployment, and evidence for funding.

To pilot this approach, a Translational Impact Measurement Consortium could be launched—starting with a small coalition of academic, investment, and public stakeholders committed to developing practical metrics. Such a consortium would function most effectively when situated within—and sustained by—an active ecosystem and community.

8. Implications for Policy, Investment, and Research Design

Adopting a translation-aware staged impact measurement framework has several implications:

- **For universities:** a structured way to articulate real-world relevance without compromising academic independence.
- **For funders and foundations:** earlier entry points for strategic funding aligned with learning and experimentation.
- **For researchers and think tanks:** a flexible structure that enables empirical research, prototyping, and tool development across academic, policy, and capital systems.

More broadly, it reframes impact measurement from a retrospective accounting exercise into a forward-looking coordination mechanism.

9. Global Impact of Applied Research

A number of impact case studies focus on the University of Cambridge's collaborations and spin-outs to showcase how research can be applied to solve societal problems. Each case study is aligned to United Nations' Sustainable Development Goals (SDGs). Examples of influential entrepreneurial engines from US universities were kindly suggested by Patrik Schmidle from CARI Health.

We are developing a table to characterise each case study against the structured dimensions of this framework.

9.1 University of Cambridge's Impact Case Studies

- **Transforming infrastructure through smart data:** University of Cambridge, The Alan Turing Institute and industry (SDGs 9 & 11). On 15 July 2021, the world's first 3D-printed steel bridge was unveiled in Amsterdam by Queen Máxima of the Netherlands. The 12m-long stainless steel bridge, for the use of pedestrians and cyclists, was built by robots and printed by Dutch company MX3D. Research on sensor and digital twin technologies were applied to monitor the bridge's structural health. The build was supported by the Lloyd's Register Foundation[1].
- **From research group to company providing equitable medicine** (SDG 3). Based at the Cambridge's Wellcome Genome Campus, Congenica is a university's spin-out. It was built upon the work of research teams led by Prof Richard Durbin and Prof Matthew Hurles, who were involved in the 10,000 UK Genome Project. They reached out to Nick Lench to translate the benefits of clinical sequencing into practical applications. From an early focus on rare diseases, application areas include somatic oncology and pathogen surveillance to prevent infectious diseases, crucial in the post-pandemic landscape[2].
- **CardiaTec: the heart of AI and research innovation** (SDG 3). A university spin-out led by Prof Namshik Han, Raphael Peralta and Thelma Zablocki, Cardiatec is underpinned by decades of computational drug discovery and industry expertise. CardiaTec is accelerating its impact by applying artificial intelligence on large-scale multi-omic datasets: genome, proteome, transcriptome, epigenome, and microbiome data. It collaborates with GSK, AstraZeneca and Storm Therapeutics, and drug discovery partners like LifeArc and Cancer Research Horizons/CRUK Cambridge Centre[3].
- **Precision medicine in cancer treatment** (SDG 3). Vector Bioscience Cambridge is a university spin-out from the Cambridge Judge Business School Accelerator, with Lluna Gallego as CEO and co-founder. The aim is to provide precision medicine by bringing the 3 Ls to life: low side effects, localised delivery

and loading (greater uptake of drugs). The company collaborates with UK and international researchers[4].

- **Creasallis: an award-winning creative antibody solutions startup** (SDG 3). Creasallis spun off the university's Accelerate@Babraham programme with Dr Zahra Jawad as CEO and founder. It provides new protein engineering solutions for antibody-based therapeutics offering enhanced efficacy, safety and lower drug costs. Its debut technology, CreaTap, is a simple solution to enhance the penetration of antibody-based drugs into the tumour. In 2023 it was celebrated as the Cambridgeshire Startup of the Year. It has received international investment and is growing steadily[5].
- **Boosting the efficiency of clean solar energy** (SDG 7). Cambridge Photon Technology (CPT) is another university spin-out, funded to commercialise renewable energy research. CPT developed a patented photon-multiplier technology that allows silicon solar panels to generate more power by converting wasted sunlight into usable energy. Its solution fits into standard solar modules without any redesign or expensive manufacturing changes. The company has attracted investment by Cambridge Enterprise Ventures, Spectrum Impact, Tybourne Capital, Providence Investment Company and SourceSquared.[6].

9.2 Entrepreneurial Engines in US Universities: UCSD, SDSU, Stanford and UC Berkeley

Patrik Schmidle from CARI Health acknowledges the support of the University of California, San Diego UCSD programmes for his company, particularly with the transition from lab to market. He highlighted the following initiatives.

- **Qualcomm Institute (QI)** is a nonprofit research organisation, dedicated to bringing together interdisciplinary teams at the university and their partners. It leverages cutting-edge technology to address real-world wicked problems. QI focuses on technology, culture, energy, the environment, and health. [7].
- **The Institute for the Global Entrepreneur (IGE)** focuses on the acceleration of medical innovation, from the bench to market. It provides support with industry mentors, executive advisors, investors and strategic corporate partners. IGE offers UCSD clinical test facilities and clinical trial design services. [8].
- **TechStars** is a privately held global, mentorship-driven accelerator that partnered with San Diego State University (SDSU). It supports founders, mentors, investors and partners. Since 2006 they have helped launch and grow thousands of companies, including 22 unicorns [9].
- **Brink SBDC** is funded through a Cooperative Agreement with the US Small Business Administration and a grant from the California Office of the Small Business Advocate, being a specialty centre of the San Diego University and the Imperial Small Business Development Center Network. SBDC has been ranked number one by the San Diego Business Journal. [10].

- **Stanford's Biodesign** aims to advance healthcare and health equity through innovation, education, translation and policy engagement. It promotes collaboration and values diversity by making Biodesign opportunities accessible to individuals from all backgrounds and geographies. Initiatives like the DxD Pathways in Health Technology program have been created to spark interest in health innovation among underrepresented groups [11].
- **University of California Berkeley's SkyDeck** was formed as a partnership between UC Berkeley's Haas School of Business, the College of Engineering and the Office of the Vice Chancellor for Research. It offers a nurturing environment for startups to grow and launch. Its ecosystem includes a network of advisors, industry partners and investors [12].

10. Conclusion and Future Research Agenda

This article argues that the persistent disconnect between universities, capital, and real-world projects is not inevitable. Rather, it reflects the absence of translational infrastructures capable of aligning diverse forms of value and evidence. By proposing a staged framework for impact measurement tailored to university-originated outputs, this paper seeks to open a research agenda at the intersection of education, investment, and societal transformation.

Our case studies show that applications of research can bring global benefits by aligning to UN's sustainable development goals. Many universities are striving to commercialise their research as impact is now required from funders. The upcoming REF2029, the Research Excellence Framework in the UK, will give more weight to research case studies from faculties and departments that demonstrate societal impact, policy changes, and inclusive innovation.

Future work may empirically test this framework through funded pilot programs, comparative case studies, and cross-institutional collaborations. In particular, the context-embedded measurement architecture and scorecard generation process offer a concrete research pathway: beginning with the structurally characterised case studies as seed data, accumulating network-embedded impact assessments across a growing corpus of translational investments, and progressively extracting the typified archetypes from which empirically grounded, portable decision tools can be derived.

Links to startups and accelerators

[1] www.smartinfrastructure.eng.cam.ac.uk/news/worlds-first-ever-3d-printed-steel-bridge-installed-amsterdam

[2] www.congenica.com/

[3] https://cardiatec.ai/

[4] https://vectorbiocam.com/

[5] https://creasallis.com/

[6] www.cambridgephoton.com/

[7] https://qi.ucsd.edu/services/qualcomm-institute-innovation-space/

[8] https://ige.ucsd.edu/

[9] https://www.techstars.com/blog/san-diego-state-university-accelerator

[10] https://www.sandiego.edu/sbdc/

[11] https://biodesign.stanford.edu/

[12] https://skydeck.berkeley.edu/

References for current and further versions

Berghäuser, H., Krauter, T., Stolz, C., Komaromi, P., & Butz, B. (2025). Mapping knowledge transfer of German universities – an analysis of 145 mission statements. *European Journal of Higher Education*, *15*(sup1). doi.org

Castañeda, D. I., & Cuellar, S. (2020). Knowledge sharing and innovation: A systematic review. *Knowledge and Process Management*, *27*(3), 159–173.

Cunningham, J. A., Menter, M., & Starke, F. (2025). The evolution of university technology transfer research: A text mining approach. *The Journal of Technology Transfer*, *50*(3), 1231–1268. https://doi.org/10.1007/s10961-024-10133-2

Hassanzadeh, M. (2025). Knowledge translation: A mechanism for dialogue between the university and society. *Sciences and Techniques of Information Management*, *11*(1), 1–9.

Loh, C. E., Lim, F. V., & Ow, R. F. L. (2025). Digital crossings: A case study of a knowledge mobilisation approach for translating research into practice. *Discourse, Context & Media*, 66, Article 100910. https://doi.org/10.1016/j.dcm.2025.100910

Lundmark, C., Nilsson, J., & Krook-Riekkola, A. (2026). Taking stock of knowledge transfer studies: Finding ways forward. *Sustainability*, *18*(4), 1102.

Marzocchi, C., Kitagawa, F., & Sánchez-Barrioluengo, M. (2020). Reconceptualising knowledge exchange and higher education institutions: Broadening our understanding of motivations, channels and stakeholders. *Studies in Higher Education*, *45*(10), 2003–2014.

Newman, J. (2025). Managing and incentivising research impact: Evidence from Australia. *Higher Education Policy*, *38*(2), 365–385.

Thune, T., Reymert, I., Gulbrandsen, M., & Simensen, E. O. (2023). Populating the science-policy co-production space: Academic and policymaker perspectives on knowledge exchange. *Studies in Higher Education*, *48*(5), 733–746. https://doi.org/10.1080/03075079.2023.2187772.

Ulrichsen, T. C., & Athanassopoulou, N. (2023). *Commercialising social science research: Insights from the University of Cambridge on key barriers, enablers and pathways to success*. University of Cambridge.

Ulrichsen, T. C., & Kelleher, L. (2025). *University knowledge exchange for economic growth: A framework for guiding understanding and evidencing success*. Policy Evidence Unit for University Commercialisation and Innovation, University of Cambridge. https://doi.org/10.17863/CAM.122918

Whiteley, A., Haigh, K., & Wake, D. (2020). *Universities and colleges and the Industrial Strategy: Exploring data on knowledge exchange, research and skills*. Industrial Strategy Council. universitiesuk.ac.uk